

	DEPARTMENT OF MECHATRONICS AND INDUSTRIAL ENGINEERING			(B)
	ODD semester TEST (1) Academic year 2023/2024			
	Subject	INTRODUCTION TO ENGINEERING SCIENCE		
	Class	B2MI/E & B4MI/E	Date	27 / 12 /2023
	Max. Marks	40	Time	1 hours
Student Name			Class	
			Reg. No	

Question No.	Q ₁	Q ₂	Q ₃	Total
Maximum Marks	20	20	20	TOTAL Out of 40
Marks Scored				
Lecturer	Dr. Fawaz Ahmed Ghaleb		Signature	

NOTES: Answer **Question No. (1)** and any other **ONE** questions.

QUESTION (1) (20 Marks)

Fill the following statement

- Engineers play a significant role in the design and maintenance of a nation's infrastructure, including systems,, and transportation.
- Engineers consider important factors such as and when designing these products.
- Good engineers have time-management skills that enable them toand
-is a broad discipline that encompasses many different scientific principles and associated mathematics that underlie engineering.
- Engineering Science aims the to provide a comprehensive grounding in relevant to engineers by an ability to identify, formulate, and solve engineering problems by applying principles of engineering,, and mathematics.
- Engineers are problem solvers. Once they have obtained a solution to a problem, they need toeffectively their solution to various..... inside or outside their organization.
- The title of a report should beinformative description of the
- Mechatronics & and industrial engineering developed in direct response to industrial demand for engineers with, is a combination of, control,and systems design engineering streams.

Introduction to engineering science

Test (1) Model (B)

Manor Gijoshi.

Q1:- Fill the following statement:-

- ① Communication systems, public utilities, and transportation.
- ② sustainability, efficiency, cost, reliability and safety
- ③ work productively and efficiently.
- ④ Engineering science
- ⑤ Science, Complex, Science
- ⑥ Communicate, people
- ⑦ a brief, report contents
- ⑧ multi-disciplinary skills, Mechanical, electronics, control, Computer and ---

Q2:- Match part I and II

- 1 - F
- 2 - E
- 3 - H
- 4 - D
- 5 - C
- 6 - A
- 7 - J
- 8 - I
- 9 - B
- 10 - G

QUESTION (2) (20 Marks)**Match part I and II**

I								II		
1-	It is another way of conveying information in a brief way to interested individuals							A	Executive summaries	
2-	To provide the reader with copies of all original data sheets, diagrams, and supplementary notes							B	Electrical Engineering	
3-	An ability to communicate effectively with a range of audiences							C	Abstract	
4-	It is a broad discipline that encompasses many different scientific principles and associated mathematics that underlie engineering.							D	Engineering Science	
5-	It is a very important part of a report							E	Appendix	
6-	Are means of communicating to people in top management positions, such as a vice president of a company, the findings of a detailed study or a proposal.							F	Short memos	
7-	It is a combination of mechanical, electronics, control, computer and systems design engineering streams.							G	Civil Engineering	
8-	It is the body of knowledge that explores the physical and natural world.							H	Engineering Science aims	
9-	It is branches including power generation, power transmission and distribution, and controls.							I	Science	
10-	It is including structural, environmental, transportation, water resources, and geotechnical.							J	Mechatronics & Industrial Engineering	
I	1	2	3	4	5	6	7	8	9	10
II										

QUESTION (3) (20 Marks)**Explain briefly why?**

1. Why is basic science important in engineering?

.....

.....

2. Why is technical report important in engineering?

.....

.....

3. Why is an abstract important to the readers?

.....

.....

4. Why are good oral and written communication skills important in engineers.

.....

.....

5. Why is mechatronics & industrial engineering department being popular?

.....

.....

②③:- Explain briefly why?

① For understanding how the world works and developing new technologies

② Because It's dealing with experimental investigations.

③ Because readers often read this part first to decide if they should read the report or not.

④ Because Communicate your solution to a problem is as important as the solution itself. Cause the results of all your effort may not be understood and appreciated if you cannot communicate effectively.

⑤ Because of the wide and equal opportunities work in the other areas such as electrical, systems, Computer engineering, etc.